

HW5 (50 Points)

In Zybooks, you had covered some “Regression” techniques like Linear Regression, PCA etc. Now it’s time to take this a step forward. For this HW, you will implement the Classification, Regression techniques using “Weka”. WEKA is a powerful tool in Datamining that comes with various pre-built algorithms and no coding is needed for the GUI version that will be used for this project. The goal is not to overburden you with new algorithms but to make you appreciate the various techniques used in the “Big Data” world by combining Datamining and Machine Learning. Also, there are plethora of tutorials about WEKA and its uses. Please feel free to use them.

- For the final document, please provide screenshots and description for every step of the task and submit your file in the following format: **“Firstname_Lastname_HW5.pdf”**.

Task 1) Using WEKA: (50 Points)

The goal of his task is to introduce you to the concepts of WEKA. In order to complete this task successfully, please do the following:

- Download and Install WEKA.
- Go to the folder that contains the “Preinstalled Datasets”.

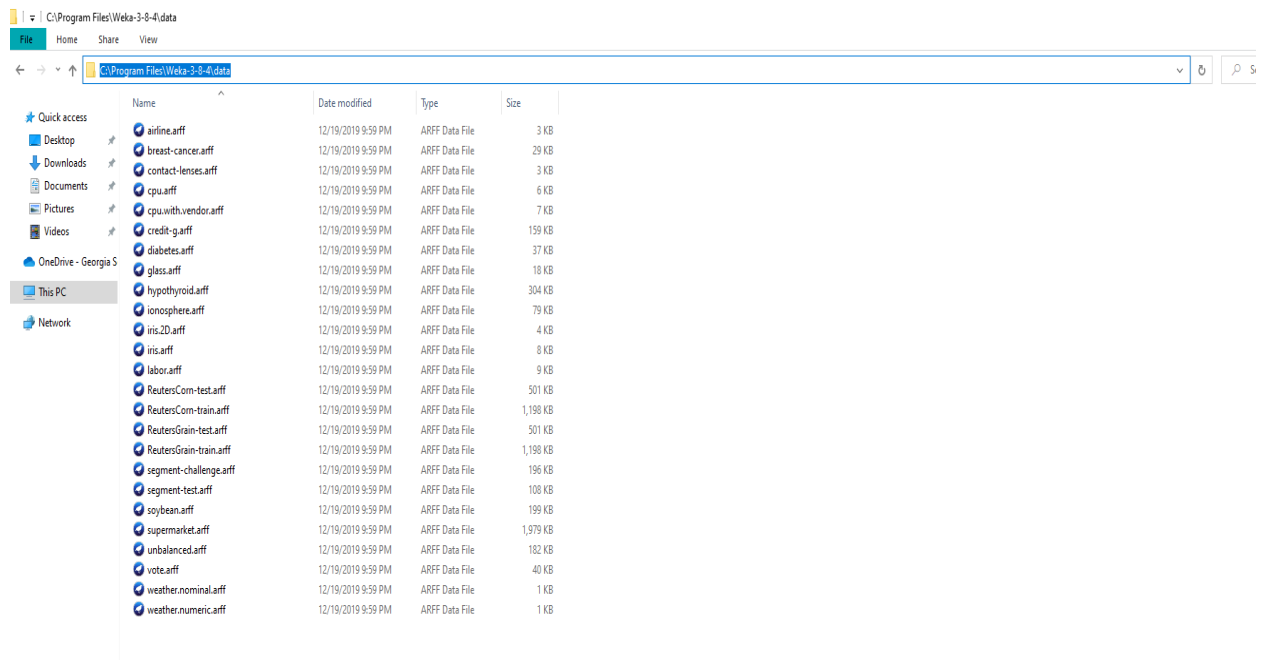


Fig 1. Sample Datasets from WEKA

- Use any **2 data sets** from the given list and use it in WEKA and apply the following classification and regression algorithms on each and show the output. The main focus should be to predict the best algorithm from the list for that particular dataset. Please remember that not all Algorithms work for all the data sets.
 1. **Naïve Bayes**
 2. **Logistic Regression**
 3. **SMO (Support Vector Machines)**
 4. **K-Nearest Neighbor**
 5. **J48 (Decision Tree)**
 6. **Random Forest**

For each algorithm mentioned above, please do the following:

- 1) Load the Dataset into WEKA.
- 2) Go to the “**Classify**” tab.
- 3) Test the options using “**Use Training Set, Cross Validation with 10 folds, Percentage Split (80%)**”.
- 4) From the classifier output window, grab a screenshot (as shown in Fig. 2) of each test option for each test and name the figure with the Algorithm and the test type. E.g.: “**Naïve Bayes, Use Training Set**”.
- 5) At the end, list which column(s) from “**Detailed Accuracy By Class**” shown in the figure below lists the best algorithm for each dataset? Also, for each data set please list the best algorithm?

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.000	0.000	?	0.000	?	?	0.778	0.041	Active
	1.000	1.000	0.988	1.000	0.994	?	0.778	0.996	Inactive
Weighted Avg.	0.988	0.988	?	0.988	?	?	0.778	0.985	

=== Confusion Matrix ===

```

a  b  <-- classified as
0  2 |  a = Active
0 169 | b = Inactive

```

Fig. 2. Classifier Output